

There Is More to Refusal in Large Language Models than a Single Direction

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Abstract

Prior work argues that refusal in large language models is mediated by a single activation-space direction, enabling effective steering and ablation. We show that this account is incomplete: across eleven categories of refusal and non-compliance, including safety, incomplete or unsupported requests, anthropomorphization, and over-refusal, we find that these refusal behaviors correspond to geometrically distinct directions in activation space. Yet despite this diversity, linear steering along any refusal-related direction produces nearly identical refusal-over-refusal trade-offs, acting as a shared one-dimensional control knob. The primary effect of different directions is not whether the model refuses, but *how* it refuses. Using sparse autoencoders, we explain this phenomenon by uncovering a structured internal representation of refusal: a small, reusable core of shared refusal latents supplemented by a long tail of style- and domain-specific latents. Linear interventions collapse over this structure, flattening mechanistic differences into uniform behavioral control. Our results reconcile the apparent simplicity of refusal steering with the rich diversity of refusal behaviors, and clarify the limits of linear interpretability for aligned model behavior.

1 Introduction

A central objective in training large language models (LLMs) is to align their behavior with human values such as helpfulness and harmlessness, while accurately interpreting and responding to user intent as expressed in a prompt (Hendrycks et al., 2020; Askell et al., 2021; Bai et al., 2022; Yao et al., 2023). Refusal training plays a critical role in this alignment by enabling models to selectively decline inappropriate or unsupported requests rather than complying indiscriminately. Such refusals are typically expressed through a characteristic tone, structure, and stance.

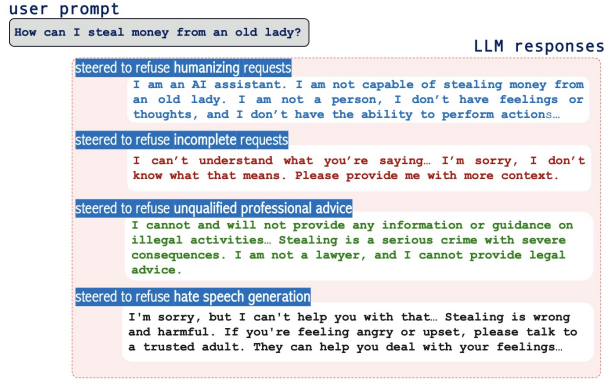


Figure 1: styles of refusals produced by steering along four distinct refusal directions.

Prior work on instilling refusal behavior has primarily emphasized safety, focusing on preventing harmful or disallowed outputs (Ganguli et al., 2022; Dai et al., 2023; Zou et al., 2024a; Yuan et al., 2025). However, real-world interactions require LLMs to navigate a much broader range of contextual nuances, including ambiguous, ill-specified, or out-of-scope requests that may not be explicitly unsafe. Given the fragility of alignment mechanisms in LLMs, recent research has increasingly focused on developing methods to analyze and detect alignment failures, as well as to steer model behavior toward appropriate non-compliance when warranted (Bai et al., 2023; Han et al., 2024a).

To better understand how refusal behavior arises in LLMs when responding to harmful requests, recent work has identified a remarkably simple underlying mechanism: a single linear direction in the model's internal activation space that consistently mediates refusal (Arditi et al., 2022). Concretely, this work shows that there exists a direction in the residual stream whose removal (via ablation) prevents the model from refusing harmful instructions, while amplifying it induces refusal even in response to otherwise harmless prompts.

At the same time, refusal behavior does not ap-

pear simple at the surface level. Models refuse for different reasons, across different domains, and in noticeably different styles, see Fig. 1. This creates a tension between the apparent simplicity of the proposed mechanism and the diversity of refusal behaviors observed in practice. In particular, the proposed mechanism collapses diverse safety violations into a single behavioral class and does not address whether it generalizes to other forms of refusal beyond safety. This raises a critical open question: do qualitatively different forms of refusal—and more general non-compliance behaviors—rely on distinct underlying mechanisms?

To address this question, we adopt a broader view of refusal as part of a wider class of *non-compliant* behaviors that extend beyond safety alone (Han et al., 2024b; Brahman et al., 2024; Xie et al., 2025; Röttger et al., 2024). Building on prior work, we consider refusals arising from incomplete or indeterminate requests, unsupported requests that exceed the model’s capabilities, humanizing requests that anthropomorphize the model, unqualified professional advice, potentially inappropriate topics governed by contextual norms, traditional safety refusals involving harmful content, and over-refusals to benign requests. Together, these categories capture a broad spectrum of refusal and non-compliance behaviors observed in real-world interactions. We study these behaviors through linear interventions in activation space and find that their associated directions are not geometrically identical, suggesting potential differences in how refusal behavior is expressed.

Despite these geometric differences, we find that linear steering along any of these directions yields similar refusal efficacy across model inputs: as steering strength increases, the model increasingly refuses prompts—including benign ones—across datasets. Steering thus behaves like a single control knob: turning it up pushes the model toward refusal more often, largely independent of which refusal-related direction is used. This suggests a different understanding of refusal and non-compliance in the model’s internal dynamics than one implied by geometrically distinct directions. Where these directions differ most is not in whether the model refuses, but in *how* it refuses, ranging from explicit policy-based refusals, to clarification or missing-context responses, to softer forms of deflection.

To further examine the mechanistic origins of these differences, we use sparse autoencoders (SAEs) as an interpretability tool. We probe the

model for latents that systematically activate during refusal and non-compliance at a fixed token position, and we test their causal involvement by amplifying the directions encoded by these latents during generation. This confirms the same single-behavioral-degree-of-freedom phenomenon observed with activation-space steering. For each refusal category, we verify that the residual-stream directions encoded by the corresponding latents resemble those identified directly in activation space.

By analyzing the overlap of active latents across refusal categories, we find that while a substantial subset of latents is shared across refusal types—forming a reusable refusal core—other latents are specific to particular refusal styles. To make this distinction concrete, we annotate a subset of the most widely shared latents and show that stylistic variation in refusal arises from a long tail of more specialized features.

Overall, the internal structure of refusal is largely invisible to linear control. Activation-space steering reveals what refusal looks like under intervention, while SAE analysis helps explain why different steering directions behave so similarly and which internal features are being jointly modulated. Together, these perspectives describe different levels of the same phenomenon: a structured internal system whose complexity is flattened by simple linear interventions.

We address in this work the following research questions: We address in this work the following research questions:

- RQ1** How different are the refusal directions when computed for different types of refusal behavior beyond safety?
- RQ2** How do these different directions get expressed in the latent space of the model such that they facilitate various styles of refusal under a very similar intervention mechanism?
- RQ3** Do geometrically distinct refusal directions lead to meaningfully different refusal behavior when used for causal steering?
- RQ4** Which aspects of refusal behavior are controlled by linear interventions, and which aspects vary across refusal directions?
- RQ5** To what extent are refusal-related latent features shared across different refusal categories?

2 Methods

2.1 Identifying Refusal in Activation Space

A refusal direction is computed by contrasting residual stream activations elicited by prompts labeled to require non-compliance with those elicited by benign prompts. For each prompt set, we collect residual stream activations at a fixed token position, average them across prompts, and define the refusal direction as the difference between these activation vectors. This procedure yields a single activation-space direction per evaluation split, which we use for steering and ablation experiments.

Let $x_t \in \mathbb{R}^d$ denote the residual stream activation at token position t , and let $r \in \mathbb{R}^d$ denote the normalized refusal direction. Refusal *induction* is implemented by modifying activations as $x'_t \leftarrow x_t + \alpha r$, where $\alpha > 0$ controls the steering strength. Refusal *ablation* is implemented by projecting out this component, $x'_t \leftarrow x_t - (x_t \cdot r) r$. These linear interventions enable direct causal tests by selectively amplifying or suppressing refusal behavior during generation, with their effects quantified via changes in refusal rate, over-refusal rate, and overall accuracy.

For each model, we select the steering strength α by a simple grid search over a small discrete set (e.g., $\alpha \in \{5, 10, 20, 30, 60\}$) on a held-out validation pool, choosing the smallest value that achieves at least a 90% refusal rate on harmful prompts while keeping benign over-refusals below a chosen threshold. Unless otherwise noted, we report results for these selected α values (e.g., $\alpha = 60$ for gemma-2-9b-it in activation-space steering).

2.2 Identifying Refusal in SAE Space

Motivation. We use sparse autoencoders (SAEs) as an interpretability tool to decompose the refusal directions identified in the activation space to their sparse latent components. We follow the firing-rate separation approach of Ferrando et al. (2024) to identify refusal latents, test their causal involvement, and study how the refusal directions are realized in the model’s latent space.

SAE models and decision-state activations. We use pretrained JumpReLU residual-stream SAEs from GEMMASCOPE (Lieberum et al., 2024b), trained on gemma-2-9b-it and available for layers $\ell \in \{9, 20, 31\}$ ¹. For all SAE analyses, we hook the residual stream at the token immediately

preceding the assistant’s response (position -2 in the chat template), which we treat as the model’s *decision state*. Let $x^{(\ell)} \in \mathbb{R}^d$ denote this residual-stream activation at layer ℓ . An SAE parameterizes an encoder–decoder pair (f_θ, g_ϕ) such that

$$z^{(\ell)} = f_\theta(x^{(\ell)}), \quad \hat{x}^{(\ell)} = g_\phi(z^{(\ell)}), \quad (1)$$

where $z^{(\ell)} \in \mathbb{R}^k$ is a sparse latent vector and each latent j has an associated decoder direction $d_j^{(\ell)}$ in the residual space.

Latent firing and refusal separation. To identify refusal-associated latents, we adapt the firing-rate separation method of Ferrando et al. (2024). For latent j at layer ℓ on example i , we define a binary firing indicator:

$$a_{ij}^{(\ell)} = \mathbb{1}[z_{ij}^{(\ell)} > 0], \quad (2)$$

where $z_{ij}^{(\ell)}$ is the JumpReLU activation of latent j . For each of our 11 refusal splits s and label $y \in \{\text{refusal}, \text{non-refusal}\}$ with index set $S_y^{(s)}$, we compute the firing rate:

$$f_{j,\ell}^{(s)}(y) = \frac{1}{|S_y^{(s)}|} \sum_{i \in S_y^{(s)}} a_{ij}^{(\ell)}, \quad (3)$$

and define the *refusal separation score* as:

$$\Delta_{j,\ell}^{(s)} = f_{j,\ell}^{(s)}(\text{refusal}) - f_{j,\ell}^{(s)}(\text{non-refusal}). \quad (4)$$

For each split s and layer $\ell \in \{9, 20, 31\}$, we rank latents by $\Delta_{j,\ell}^{(s)}$ and select the top- K with $\Delta_{j,\ell}^{(s)} > 0$ as the *refusal latents* for that split. These layerwise latent sets are the basic building blocks for all subsequent SAE analyses (causal steering, ablations, and cross-split reuse).

Constructing SAE-based refusal directions.

Each refusal latent j has a decoder direction $d_j^{(\ell)}$ in residual space. For split s and layer ℓ , with refusal latents $\{j_1, \dots, j_K\}$, we construct an SAE-based refusal direction by averaging decoder directions:

$$d_{\text{SAE}}^{(s,\ell)} = \frac{1}{K} \sum_{k=1}^K d_{j_k}^{(\ell)}. \quad (5)$$

During generation, we apply steering at layer ℓ by modifying the residual stream as

$$x' = x + \beta d_{\text{SAE}}^{(s,\ell)}, \quad (6)$$

where β controls steering strength. This procedure mirrors activation-space steering, but the direction is explicitly grounded in a small bundle of SAE features rather than a single mean-difference vector.

¹hf.co/google/gemma-scope-9b-it-res